

What did you discover?

Then, I realised that I must have been reading the ball's direction and turning as my knees bent after my split step landing. In other words, the landing was not instantaneous. It had two parts: the touchdown, which was instantaneous, and the knee bend after landing, which was not instantaneous. So, I began timing the duration of the after-landing knee bend on slow-motion videos of high-level players. The duration was roughly 0.2 seconds.

I checked the reaction time to visual cues in humans. Reaction times were typically 0.2 to 0.25 seconds and could be improved by training. Not having time to read the ball's direction did not seem to be why players were not landing their split step precisely at the instant their opponent hit the ball.

You solved one part of this mystery. What was the next piece of this puzzle?

So why weren't the best tennis players in the world touching down their split steps at the instant their opponent hit the ball and reading the ball's direction as their knees bent after landing? This would enable them to initiate their first move towards the oncoming ball at roughly 0.2 seconds after the opponent hit it, which is sooner than any of them seem to be doing it.

I concluded, with considerable misgivings, that the best players in the world were making an important tactical mistake with their footwork.

What was that?

Additional slow-motion videos of world-class players approaching the net gave me some relief. I watched them make the same blunder time after time! They approached the net on a path that — although they did not know it yet — would have allowed them to hit a volley after one step and a reach, and the volley would often have been a winner.

Instead, here's what they did:

❶ They would hop into the air at roughly the instant their opponent hit the passing shot.



❷ They would read the direction of the passing shot and would land on the foot they would push off for their first move towards the ball. This foot would typically be the one farther from the direction they wanted to move (they would be doing a crossover step).

❸ For some reason, perhaps instinct, they would push this foot farther out, by a foot or more, away from the direction they wanted to move.

❹ This would leave them with such a wide stance that they would have to pull their other foot in, also away from their intended direction, in order to get their centre of mass over that

Matter of seconds: A split step landing has two parts: the touchdown and the knee bend. Executing it perfectly can determine how well players can return shots from their opponents.

other foot to use gravity to help them accelerate towards the ball.

❺ Because of the lateness of their landing, the duration of these two steps, and the fact that they had moved away from their intended direction, they would be unable to even make a play on the passing shot. This flawed footwork did not always occur, but when it did, the